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Journal Publications and Preprints

Physics (†authors listed alphabetically)

1. **V. Lochab** and F. Denef. Higher-loop vacuum integrals on S^D. arXiv:2610.30489 (2026).
†2. J. Bonifacio, K. Hinterbichler, Y. T. Albert Law, and **V. Lochab**. De Sitter observers and an effective field theory of horizon edge modes. arXiv:2610.28719v1 (2026).
†3. Y. T. Albert Law and **V. Lochab**. Horizon edge partition functions in $\Lambda > 0$ quantum gravity. *Physical Review Letters* **136**, 151601 (2026). arXiv:2603.20913.
†4. Y. T. Albert Law and **V. Lochab**. Gravitons on Nariai edges. *Physical Review D* **113**, 086006 (2026). arXiv:2506.02142.
5. X. Zheng, J. Dolde, **V. Lochab**, B. N. Merriman, H. Li, and S. Kolkowitz. Differential clock comparisons with a multiplexed optical lattice clock. *Nature* **602**, 425–430 (2022).
6. **V. Lochab**. Three-loop melon on spheres. ***In preparation.***
7. **V. Lochab**. A finite field approach to higher-loop vacuum integrals on spheres. ***In preparation.***
†8. M. Grewal, Y. T. A. Law, and **V. Lochab**. Spinning thermal static patch correlators. ***In preparation.***
9. **V. Lochab**. A Lorentzian interpretation of the Harish-Chandra character for Dirac fields. ***In preparation.***

Fluid Dynamics at the Microscale

10. **V. Lochab**, E. D. Ewim, and S. Prakash. Continuous flow microfluidics for colloidal particle assembly on porous substrates. *Soft Matter* **19**, 2564–2569 (2023).
11. **V. Lochab** and S. Prakash. Combined electrokinetic and shear flows control colloidal particle distribution across microchannel cross-sections. *Soft Matter* **17**, 611–620 (2021).
12. **V. Lochab**, A. Yee, M. Yoda, A. T. Conlisk, and S. Prakash. Dynamics of colloidal particles in microchannels under combined pressure and electric potential gradients. *Microfluidics and Nanofluidics* **23**, 134 (2019).

Microbiology and Physiology

13. **V. Lochab**, T. H. Jones, D. H. Dusane, C. W. Peters, P. Stoodley, D. J. Wozniak, V. V. Subramaniam, and S. Prakash. Ultrastructure imaging of *Pseudomonas aeruginosa* lawn biofilms and eradication of the tobramycin-resistant variants under *in vitro* electroceutical treatment. *Scientific Reports* **10**, 9879 (2020).
14. **V. Lochab**, T. H. Jones, E. Alkandry, J. D. West, M. H. Abdel-Rahman, V. V. Subramaniam, and S. Prakash. Evaluation of electrical properties of *ex vivo* human hepatic tissue with metastatic colorectal cancer. *Physiological Measurement* **41**, 085005 (2020).
15. R. Heald, M. Bennett, V. V. Subramaniam, D. Dusane, **V. Lochab**, P. M. Sundaram, S. Salyer, J. D. West, P. Stoodley, and S. Prakash. Printed electroceutical dressings for the inhibition of biofilms and treatment of chronic wounds. *Journal of Microelectromechanical Systems* **29**, 918–923 (2020).
16. D. H. Dusane*, **V. Lochab***, T. Jones*, C. W. Peters, D. Sindeldecker, A. Das, S. Roy, C. K. Sen, V. V. Subramaniam, D. J. Wozniak, S. Prakash, and P. Stoodley. Electroceutical treatment of *Pseudomonas aeruginosa* biofilms. *Scientific Reports* **9**, 2008 (2019). **Equal Contribution*
17. S. Roy, S. Prakash, S. S. Mathew-Steiner, P. D. Ghatak, **V. Lochab**, T. H. Jones, P. M. Sundaram, G. M. Gordillo, V. V. Subramaniam, and C. K. Sen. Disposable patterned electroceutical dressing (PED-10) is safe for treatment of open clinical chronic wounds. *Advances in Wound Care* **8**, 149–159 (2019).

Conference Publications

Fluid Dynamics at the Microscale

18. **V. Lochab**, A. Yee, Y. Li, M. Yoda, A. T. Conlisk, and S. Prakash. Directed self-assembly of colloidal particles for high aspect ratio bands. *Technical Digest of the Hilton Head Workshop: A Solid-State Sensors, Actuators and Microsystems Workshop*, Hilton Head, South Carolina, pp. 265–268 (2018).
19. K. K. Rangharajan, **V. Lochab**, and S. Prakash. Desalting high salinity shale flowback water via high-flux nanofluidic evaporation-condensation. *IEEE 30th International Conference on Micro Electro Mechanical Systems (MEMS)*, Las Vegas, Nevada, pp. 60–63 (2017).
20. C. Boone, **V. Lochab**, M. Fuest, and S. Prakash. Time-dependent gating in nanofluidic channels for active electric double layer modulation for ion transport control. *Technical Digest of the Hilton Head Workshop: A Solid-State Sensors, Actuators and Microsystems Workshop*, Hilton Head, South Carolina, pp. 198–201 (2016).

Microbiology and Physiology

21. **V. Lochab**, E. Alkandry, T. H. Jones, M. Abdel-Rahman, E. W. Martin Jr., C. R. Schmidt, S. R. Abdel-Misih, M. Bloomston, V. V. Subramaniam, and S. Prakash. μ-Electrochemical impedance spectroscopy for identifying spatial heterogeneity in tumor bearing hepatic tissue. *21st International Conference on Miniaturized Systems for Chemistry and Life Sciences (μTAS)*, pp. 501–502 (2017).
22. **V. Lochab**, T. H. Jones, D. H. Dusane, C. K. Sen, S. Roy, P. Stoodley, D. Wozniak, V. V. Subramaniam, and S. Prakash. Mechanistic determination of electroceutical bactericidal effects on *Pseudomonas aeruginosa* biofilms. *21st International Conference on Miniaturized Systems for Chemistry and Life Sciences (μTAS)*, pp. 1175–1176 (2017).

Meetings & Abstracts

Physics

1. X. Zheng, J. C. Dolde, **V. Lochab**, B. N. Merriman, H. Li, and S. Kolkowitz. Measuring gravitational redshift at the centimeter scale with a multiplexed strontium optical lattice clock. *Bulletin of the American Physical Society* **66**, V01.00133 (2021).
2. J. C. Dolde, X. Zheng, **V. Lochab**, B. N. Merriman, H. Li, and S. Kolkowitz. Enhanced differential stability measurements with a multiplexed strontium optical lattice clock. *Bulletin of the American Physical Society* **66**, S03.00002 (2021).

Fluid Dynamics at the Microscale

3. S. Prakash, **V. Lochab**, and E. Ewim. Use of combined electrokinetic and Poiseuille flows to generate organized colloidal structures. *Bulletin of the American Physical Society* **68**, N18.00011 (2023).
4. S. Prakash, **V. Lochab**, and E. Ewim. Demonstrating use of continuous flow microfluidics to assemble colloidal particles on porous substrates. *Bulletin of the American Physical Society* **67**, U34.00007 (2022).
5. S. Prakash and **V. Lochab**. Colloidal migration in microchannels under combined potential and pressure gradients. *Bulletin of the American Physical Society* **65**, D15.00012 (2020).
6. S. Prakash and **V. Lochab**. Colloidal particle migration within microchannels in combined Poiseuille and electrokinetic flows. *Bulletin of the American Physical Society* **64**, H36.00002 (2019).
7. S. Prakash and **V. Lochab**. Electrohydrodynamic printing of pre-assembled colloidal particle structures. *Bulletin of the American Physical Society* **63**, G24.00003 (2018).
8. M. Yoda, A. Yee, **V. Lochab**, and S. Prakash. Colloidal band assembly in different microchannels. *Bulletin of the American Physical Society* **62**, G37.00002 (2017).